

Program Name	B.C.A	Semester	VI
Course Title	PROJECT WORK		
Course Code:	DSC18-Project	No. of Credits	06
Contact hours	12 Hours per week	Duration off SEA/Exam	3 hours
Formative Assessment Marks	50	Summative Assessment Marks	100

GUIDELINES FOR PROJECT WORKS AND EVALUATION

PROJECT GUIDELINES

Preamble: Project work is integrated into the BCA curriculum to offer students practical software development experience, focusing on software engineering principles. Throughout the final semester, students engage in all stages of the software development life cycle (SDLC), creating reliable systems. Projects address real-world issues, demanding original, application-oriented solutions. Plagiarism is strictly prohibited, with projects expected to be non-trivial and analytical. Equal team participation is encouraged, and timely completion of all project development activities is required.

GENERAL GUIDELINES TO THE INSTITUTIONS/COLLEGES

- The Project Work schedule for the Sixth semester will be announced beforehand, detailing deadlines for various project milestones. These include submission dates for the Project Proposal, Project Acceptance, Project Synopsis, Problem Analysis Document, System Design Document, Database Design, Detailed Design, Coding and Testing, Final Report, Internal Assessment exams (at least two), and Viva/ Voce assessments.
- Students are encouraged to tackle projects addressing real-life problems, either within their colleges or in industry/research and development settings, as advised by BCA project faculty. For industry projects, faculty must verify their authenticity and originality to ensure their genuine nature.
- There shall be not more than three members in a Project team.

- At least two internal assessment exams will be held to gauge students' progress during various project stages. These assessments may encompass written tests, document scrutiny, presentations, work demonstrations, group discussions, and viva-voce sessions. This multi-faceted approach ensures an objective evaluation of students' comprehension throughout their project endeavors.

PROJECT VALUATION

External and internal evaluators jointly conduct project assessments in an impartial manner. They define precise evaluation criteria through discussions. Students present their work either through live software demonstrations or, if source code confidentiality is an issue, through verified PPT presentations with company endorsements. The assessment aims to measure students' dedication and comprehension. Evaluators verify adherence to SDLC principles in project reports, taking into account their relevance in context for equitable evaluation. The primary focus is on achieving project goals and documenting SDLC compliance, ensuring thorough assessment while addressing domain-specific nuances for impartiality.

SCHEME OF VALUATION AND MARKS DISTRIBUTION

Particulars		Marks
Internal Assessment		
Progress assessment		50
Project Report Valuation: Based on the Innovativeness and utility of the project for Industry/Academic or Society (Utility) Related studies about the project (Adequacy), Project plan & implementation-target achieved/output delivered (effectiveness).		75
Viva-Voce: 25 marks		
1	Live Demonstration (Software execution) or Dry runs (Presentation of authentic screenshots or captured videos may be used to walk through complete scenarios)-consistency and completeness	15
2	Question and Answer (Oral only or Oral and written)	10
Total Marks		150

FORMAT OF PROJECT SYNOPSIS:

Synopsis is a brief outline or general view, as of a subject or written work; an abstract or a summary of the Project Work. It must be as brief (NOT MORE THAN 10 A4 sized paperpages) as is sufficient enough to explain the objective and implementation of the project that the candidate is going to take up.

The write up must adhere to the guidelines and should include the following:

- Title of the Project.
- Introduction, objectives and scope of the Project.
- Tools / Platform, Hardware and Software Requirement specifications.
- Analysis (DFDs at least up to second level, ER Diagrams/ Class Diagrams, Database Design etc, as per the project requirements).
- A complete structure which includes: Number of modules and their description to provide an estimation of the student's effort on the project, Data Structures as per the project requirements for all the modules, Process logic of each module, testing process to be used, reports generation (Mention tentative content of report).
- Whether Industry Defined/Client Defined/User Defined Project? Mention the type. Mention the name and Address of the Industry/Client.
- Limitation of the project.
- Future scope and further enhancement of the project.

GUIDELINES FOR PREPARATION OF DISSERTATION

ORGANISATION OF THE DISSERTATION

The dissertation shall be presented in a number of chapters; starting with Introduction and ending with Conclusion. Each of the chapters will have precise title reflecting the contents of the chapter. A chapter can be subdivided into sections, sub-sections and sub-sub-section so as to present the content discretely and with due emphasis.

Sequence of items in Dissertation Report

The following sequence may be followed in the preparation of the final dissertation report:

- Cover Page (On the hardbound cover)
- Title Page (Inner Cover Page)
- Certificate from the Institute
- Certificate from the Company/Industry/Institution from where the project has been completed
- Declaration Certificate (signed by both Student(s) and guide)
- Acknowledgement
- Table of Contents (with page numbers).
- List of Figures (with figure number, figure titles and page numbers)
- List of Tables with table number, table title and page number.
- Chapters 1,2,.....

Sample Chapters:

1. Introduction

- a. Introduction of the System
 - i. Project Title
 - ii. Category
 - iii. Overview
- b. Background
 - i. Introduction of the Company
 - ii. Brief note on Existing System
- c. Objectives of the System
- d. Scope of the System
- e. Structure of the System
- f. System Architecture
- g. End Users
- h. Software/Hardware used for the development
- i. Software/Hardware required for the implementation

2. SRS

- a. Introduction (Brief write-up about SRS)
- b. Overall Description
 - i. Product perspective
 - ii. Product Functions
 - iii. User characteristics.
 - iv. General constraints

- v. Assumptions
- c. Special Requirements (Software / Hardware-if any)
- d. Functional requirement.
 - i. Module 1
 - ii. Module 2
- e. Design Constraints
- f. System Attributes
- g. Other Requirements (if any)

3. System Design (Functional Design)

- a. Introduction (brief write-up about System Design)
- b. Assumptions and Constraints
- c. Functional decomposition
- d. Description of Programs
 - i. Context Flow Diagram (CFD)
 - ii. Data Flow Diagrams (DFDs–Level 0, Level 1, Level 2)
- e. Description of components
 - i. Functional component 1
 - ii. Functional component 2

4. Database Design (or Data structure)

- a. Introduction (brief write-up about Database design)
- b. Purpose and scope
- c. Table Definition
- d. ER diagram

5. Detailed Design (Logic design of modules)

- a. Introduction (brief write-up about Database design)
- b. Structure of the software package (structure chart)
- c. Modular decomposition of the System
 - i. Module1
 - 1. Inputs
 - 2. Procedural details
 - 3. File I/O interfaces
 - 4. Outputs
 - 5. Implementation aspects (if any)
 - ii. Module 2

6. Program code listing

- a. Database connection
- b. Authorization / Authentication
- c. Data store / retrieval /update
- d. Data validation
- e. Search

- f. Named procedures / functions
- g. Interfacing with external devices (if any)
- h. Passing of parameters
- i. Backup/recovery
- j. Internal documentation

7. User Interface (Screens and Reports)

- a. Login
- b. Main Screen / Home page
- c. Menu
- d. Data store / retrieval / update
- e. Validation
- f. View
- g. On screen reports
- h. Data Reports
- i. Alerts
- j. Error messages

8. Testing

- a. Introduction (brief write-up about Software Testing)
 - i. Unit Testing
 - ii. Integrate Testing
 - iii. System Testing
- b. Test Reports

- Conclusion
- Limitations
- Scope for enhancement (future scope)
- Abbreviations and Acronyms (list)
- Bibliography / References (list in specified format)

Note: Do not include any header or footer in any page of the report. Only page numbers should be mentioned at the bottom center of each page. 'n' copies of dissertation along with soft copy in CD should be prepared by the candidate.

DISSERTATION FORMAT

Paper Quality

The dissertation shall be printed on white bond paper, whiteness 95% or above, weight 70 gram or more per square meter.

Paper Size

The size of the paper shall be standard A4; height 297 mm, width 210 mm.

Type-Setting, Text Processing and Printing

The text shall be printed employing LaserJet or Inkjet printer, the text having been processed using a standard text processor. The standard font shall be Times New Roman of 12 pts with 1.5 line spacing.

Page Format

- Top margin .5"
- Bottom margin .5"
- Left margin 1"
- Right margin .75"

AUXILIARY FORMATS

- Binding: The dissertation shall be hard cover bound in leather or rexin.
- Front Covers: Full title of dissertation in 6 mm 22-point size font properly centered and positioned at the top. Full name of the candidate in 4.5 mm 15-point size font properly centered at the middle of the page. A 40 mm dia replica of the college emblem followed by the name of the Department and the year of submission, each in a separate line and properly centered and located at the bottom of the page.
- Lettering: All lettering shall be embossed in gold.
- Bound back: The degree, the name of the candidate and the year of submission shall also be embossed on the bound (side) in gold.

Program Name	B.C.A	Semester	VI
Course Title	Internship		
Course Code:	SEC-5	No. of Credits	02
Contact hours	30 Hours	Duration of SEA/Exam	---
Formative Assessment Marks	50		

GUIDELINES FOR INTRENSHIP AND EVALUVATION

Internships can cover a wide range of concepts and topics and some common concepts that can be covered under various types of internships:

Technical Skills

- Depending on the field, interns can develop technical skills such as programming languages, software tools, data analysis, design software, and more.

Soft Skills:

- Communication: Written and verbal communication skills, including effective email communication, presentations, and client interactions.
- Teamwork: Collaborating with colleagues, working in cross-functional teams, and building effective relationships.
- Time Management: Prioritizing tasks, managing deadlines, and staying organized.
- Problem Solving: Analysing challenges, identifying solutions, and making informed decisions.
- Adaptability: Handling changes, learning new processes, and adjusting to evolving situations.

Innovation and Entrepreneurship:

- Exploring innovative business ideas, product development, market research, and business model creation.

Data Analytics and Interpretation:

- Learning how to work with data, perform analysis, and derive insights to inform decision-making.

Leadership and Management:

- Developing leadership skills, understanding different management styles, and learning how to motivate teams.

Evaluation:

- Formative assessment marks (Internal Assessment - IA): Two internal tests, C1 and C2, each worth 25 marks (Internal Test and Seminar/Activity), will be conducted during the semester. Summative assessment marks will be determined by evaluating a report prepared by the student with guidance from a designated mentor in the college. The report will be submitted to the Head of the Department for assessment. Two internal faculty members will evaluate the report and submit the final sessional and summative marks to the university.

01	Prof. Shivanand S Gornale Professor of Computer Science Rani Channamma University, Belagavi	Chairman BOS in Computer Science	
02	Sri Vinaykumar V Assistant Professor of Computer Science Govt. First Grade College, Raibag	Member BOS in Computer Science	
03	Smt. Maheshwari Hiremath Assistant Professor of Computer Science Govt. First Grade College, Navabag Vijaypura	Member BOS in Computer Science	